Agney Jadav



Mechatronics Master's Student | Metal Additive Manufacturing (LPBF) · Design · Automation · Testing

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Residence permit valid until 10/2027 · EU Blue Card possible after graduation (German master's)

PROFILE

Mechatronics master's student at TH Rosenheim (thesis submitted 10/2026), available now as a working student (up to 20 h/week) or for direct entry. Thesis on rare-earth oxide inoculation of LPBF 316L stainless steel: powder-milling route, printing on a TRUMPF TruPrint 1000, metallography, porosity and hardness testing. CAD in SolidWorks (CSWA), PLC programming and digital-twin skills. Co-founder of a 3D-printing start-up that took 20+ products from CAD model to sale.

EDUCATION

10/2024 – present

Master's in Engineering Sciences – Mechatronics

Current average 2.3

Rosenheim University of Applied Sciences (TH Rosenheim), Germany

Master's thesis: Rare-earth oxide inoculation (Y_2O_3, CeO_2) of LPBF 316L stainless steel

- Developed the powder-mixing route: compared five planetary ball-milling routes (SEM, shear-cell flowability) at the FRITSCH lab; selected hardened-steel media at BPR 3:1
- Printed inoculated samples and a 316L baseline on a TRUMPF TruPrint 1000 with identical parameters; analysed porosity, hardness (HV5), etched microstructure and SEM/EDX
- Results: no significant hardness change (211.5–215.3 vs. 211.0 HV5, $p = 0.22$); 0.3 wt% inoculants cut pores > 100 µm from 13 to 0–2; 0.7 wt% Y_2O_3 caused lack of fusion; attributed yttrium cross-contamination to shared milling media (EDX)

Skills gained

- Automation & control: PLC programming (SIMATIC S7, CODESYS; ST, SFC), OPC UA, MQTT, SCADA/HMI, IIoT; MATLAB/Simulink/Stateflow, rapid control prototyping with dSPACE
- Production & design: automated-system planning and commissioning, risk assessment (ISO 12100), process capability and OEE, Lean (Toyota Production System), DFMA, statistical tolerance analysis, FEM, DfAM
- Materials & data: physical metallurgy (solidification, phase transformations, heat treatment of steel), polymer fillers and compounding, statistical testing, reliability analysis

06/2019 – 05/2023

Bachelor of Technology – Mechatronics Engineering

Note 1,46

U.V. Patel College of Engineering, Ganpat University, India

- Major project (bachelor thesis): ARISTO XT robotic arm
- Key subjects: Robotics, Control Engineering, Hydraulics & Pneumatics, Sensor & Embedded Systems, Industrial IoT, Machine Vision, CAD, Metrology

WORK EXPERIENCE

06/2026 – present

Leisure Group Leader (Freizeitbetreuer), part-time

DID, Germany

- Organise and lead weekly day trips for groups of 4–14 participants – all communication in German

06/2023 – 12/2024

Product Development Engineer (Co-Founder)

Sewing3D, India (remote from 10/2024)

- Designed 20+ 3D-printed aircraft models and brought them to market on Amazon India – from CAD model to series production
- Developed design variants per product, refined them through printed prototypes and functional tests, and optimised parts for manufacturability and cost (DfM); CAD in SolidWorks, Fusion 360 and Blender

- Set up repeatable FDM and DLP production (Bambu Lab, Creality, Anycubic) focused on part quality, machine reliability and troubleshooting

06/2023 – 12/2024

Robotics Instructor, part-time
BrightCHAMPS, India (remote)

- Taught robotics, embedded systems, sensors and electronics basics to 50+ students in India, Japan and the USA
- Guided Arduino-based project builds, including troubleshooting and basic control-system concepts

07/2022 (3 weeks)

Industrial Trainee – Production Technology
RAJOO Engineers, India (industrial training during B.Tech)

- Machine shop: CNC machining on 5-axis and VMC machines, G-code basics
- Assembly: assembly of plastics extrusion machinery
- Electrical department: PLC and sensor integration in machine automation

SELECTED PROJECTS

Master's project

Digital twin of a production facility
Team project: digital twin of a production facility in AVEVA ArchestrA IDE – SCADA/HMI, process control, reliability analysis and statistics.
AVEVA ArchestrA IDE · SCADA/HMI · statistics

University project

Hydraulic manifold for LPBF
Redesigned a hydraulic manifold for metal LPBF using DfAM principles, routing internal flow channels within LPBF manufacturing constraints.
DfAM · CAD · LPBF

Master's module

3D-printed pneumatic gripper
Designed a pneumatic gripper in PTC Creo – variant design, DfM and technical documentation through to a working 3D-printed prototype.
PTC Creo · pneumatics · DfM · FDM

SKILLS

Expert	SolidWorks (CSWA), FDM & DLP 3D printing, metallography & materials testing (sample preparation, hardness testing, optical microscopy)
Proficient	LPBF (TRUMPF TruPrint 1000), MATLAB/Simulink, Arduino & microcontrollers, digital twins & process simulation
Working knowledge	AutoCAD, PTC Creo, ANSYS (FEM), PLC (SIMATIC S7, CODESYS), dSPACE, C/C++, CNC & G-code, DfAM, SPC, SEM/EDX, statistical analysis; currently learning Siemens NX

LANGUAGES

German	B1 (telc B2 exam 11/2026)
English	C1 (IELTS 8.0)
Native	Hindi, Gujarati

CERTIFICATES & AWARDS

	Certified SOLIDWORKS Associate (CSWA)
2023	Machine Design & Additive Manufacturing
2023	ROBOFEST-Gujarat 3.0 – 3rd prize, prototype stage, self-balancing robot (team prize ₹5 lakh)